Exercises

1	2	3	4	5	6

Surname, First name

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1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9
0	0	0	0	0	0	0

5EZB0 Math 2
Re-sit interim period

Particular Ans on paper exam instructions

- Write in a black or blue pen.
- You answer open-ended questions by using the text box. Provide your answers on the papers inside the answer box underneath a question. ***If you need more space for your answers, use the extra space at the end of the exam, and clearly indicate there which question you continue answering. In the text box of the particular question, clearly state that you proceed with your answer on a different page.***
- **The solutions should be formulated clearly, arranged orderly, and should include clear explanations of all steps that are needed to obtain the final answer.**
- Hand in all pages. Do not remove the staple. If you remove it anyhow, check that you hand in all pages.

Dear student,

This is the re-sit exam for 5EZB0. Write down your name and your student ID at the appropriate places above. Make sure that you enter your student ID by fully coloring the appropriate boxes. On the examination attendance card, you fill in the PDF number. You can find the correct number on the top of the first page of your exam (e.g. 1234.pdf).

Please read the following information carefully:

Date exam: 12/08/2024
Start time: 9:00
End time: 12:00 (+30 minutes for time extension students)

Number of questions: 5
Maximum number of points/distribution of points over questions: 80=20+18+10+22+10

Permitted examination aids

- Scrap paper (fully blank)
- Formula sheet, provided by the invigilator
- Calculator **(non-graphical/ non-programmable)**

Important:

- You are only permitted to visit the toilets under supervision
- Examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- The house rules must be observed during the examination
- The instructions of subject experts and invigilators must be followed
- Keep your work place as clean as possible: put pencil case and breadbox away, limit snacks and drinks
- You are not permitted to share examination aids or lend them to each other
- Do not communicate with any other person by any means

During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, smartwatch, smart glasses etc.
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- copying (in any form)
- visiting the toilet (or going outside) without permission or supervision

You can start the exam now, good luck!

Question 1
Consider the function

$f(x,y) := \ln\left(1 - \sqrt{1 - xy}\right)$

5p **1a** What is the maximal domain of definition $\mathcal{D} \subseteq \mathbb{R}^2$ of the function f? Explain your answer and sketch the domain $\mathcal{D}$.

3p **1b** Is $\mathcal{D}$ an open set, closed set, or neither? Justify your answer.

[**Note:** A domain D is *open* if for every $\mathbf{x}_0 \in D$ there exists a radius $r > 0$ such that the ball $B_r(\mathbf{x}_0)$ is fully included in D, i.e., $B_r(\mathbf{x}_0) \subset D$. On the other hand, *closed sets* are sets whose complement is open.]

3p **1c** Determine $\lim_{(x,y)\to(0,0)} f(x,y)$. Justify your answer.

6p **1d** Write down the formula for the linear approximation of the function $f(x, y)$ around a point (x_0, y_0).

Find the linear approximation of $f(x, y)$ around $x_0 = 1/e$ and $y_0 = 2 - 1/e$.

3p **1e** What is the directional derivative of the function f at $(x_0, y_0) = (1/e, 2 - 1/e)$ along the unit vector $\hat{u} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$.

Question 2

Consider the potential function defined over $\mathbb{R}^3$,

$$\varphi(x,y,z) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)^{\frac{1}{2}}$$

for constants $a, b, c > 0$.

2p **2a** What is the conservative field $\mathbf{f} = \nabla\varphi$ that corresponds to this potential?

5p **2b** What are the flow-lines corresponding to this field $\mathbf{f}$ passing through the point (x_0, y_0, z_0)?

5p **2c** A particle is travelling with constant speed $\left|\frac{d\mathbf{r}}{dt}\right| = abc$ along the flow-line of the field $\mathbf{f}$ passing through $(x_0, y_0, z_0) = \frac{1}{\sqrt{3}}(a, b, c)$. Its velocity is always pointed along the field $\mathbf{f}$. What is the rate of change of potential, i.e., $\frac{\partial \varphi}{\partial t}$, that the particle is experiencing at (x_0, y_0, z_0)?

[**Hint:** You do not need to know the flow-lines to answer this question]

6p **2d** Still for the same potential function given at the start of Question 2, fix, $a = b = 1$ and $c = 1/\sqrt{2}$. Let $\mathcal{C}$ be one revolution of the *conical helix* curve, i.e., $\mathcal{C}$ is parametrized as

$$\{(t\cos(t), t\sin(t), t) \mid 0 \leq t \leq 2\pi\}.$$

Evaluate the line integral: $\int_{\mathcal{C}} \varphi \, ds.$

Question 3

Consider the following surface (the so-called "hyperboloid of one-sheet")

$$\mathcal{S} := \left\{(x,y,z) \in \mathbb{R}^3 : x^2 - 2x + y^2 = z^2, \quad 0 \le z \le 1\right\},$$

4p **3a** Find a suitable parametrization of $\mathcal{S}$, i.e.,

$$\mathbf{r}(u,v) = x(u,v)\hat{\mathbf{i}} + y(u,v)\hat{\mathbf{j}} + z(u,v)\hat{\mathbf{k}}$$

such that $\mathcal{S}$ is described by fixed bounds on $u,\ v$. What are these fixed bounds?

[**Hint:** For instance, $y(u,v)$ could be parametrized as $y(u,v) = (1+v^2)^{\frac{1}{2}}\sin(u)$.]

6p **3b** Determine the area of the surface $\mathcal{S}$. You will have to write your answer in the following form:

Area of $\mathcal{S} = c_1 \int_0^r \sqrt{1+2t^2}dt$, where you do not have to solve the integral but write your solution in that form. You will need to find expressions for c_1, r and t, however.

[**Note**: The computation can be done both using the parametrization above, or the surface area formula of a graph $z = f(x,y)$, i.e., $\iint \sqrt{1+|\nabla f|^2}dxdy$]

Question 4

Evaluate the following integrals

6p **4a** The domain $\mathcal{R} \subset \mathbb{R}^2$ is bounded by the parabola $y = x^2$ from the bottom and the straight line $y = x + 2$ from the top. Draw $\mathcal{R}$ and determine its area by expressing it as and solving a double integral.

5p **4b** We now consider the same planar domain $\mathcal{R}$ as in Question 4a and add a third dimension to create a volume $\mathcal{V}$. Consider the domain $\mathcal{V} = \{(x, y, z) \mid (x, y) \in \mathcal{R},\ z \geq 0,\ y + z \leq 4\}$. Evaluate the integral

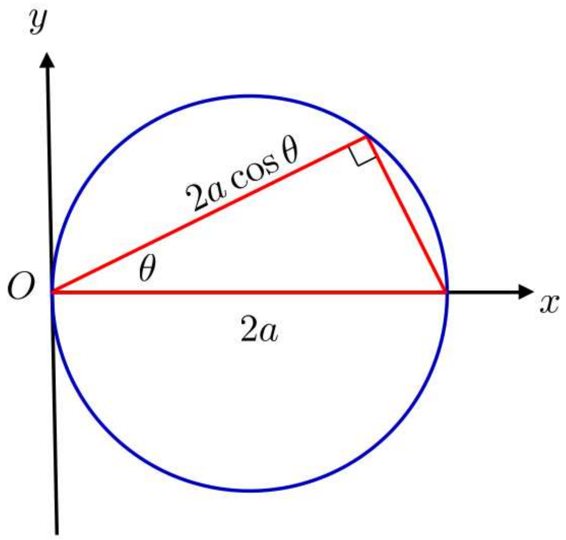
$$\iiint_{\mathcal{V}} x\, dx\, dy\, dz.$$

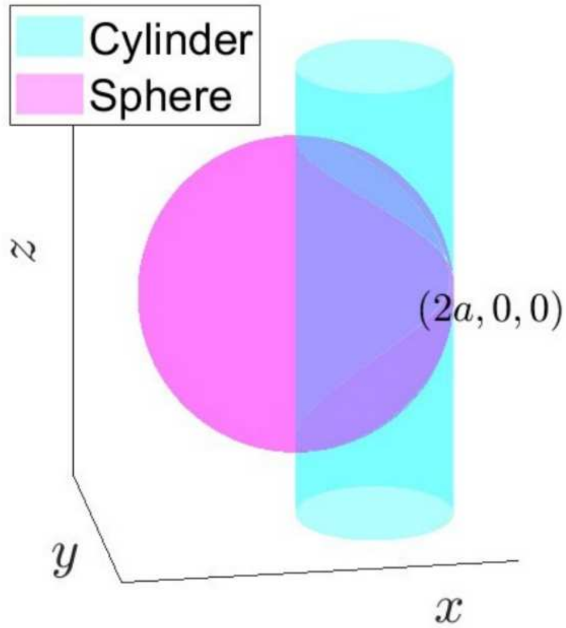
[**Note:** It is not required to reduce the final fractions]

11p **4c**

Let the domain $\mathcal{D}$ be the intersection of the sphere (centered at the origin) $x^2 + y^2 + z^2 \leq 4a^2$ and the cylinder $(x - a)^2 + y^2 \leq a^2$. Find the volume of $\mathcal{D}$.

[**Hint:** transform to a suitable coordinate system; you could use the pictures below]





Question 5

Let $\mathbf{f}(x, y, z) = (f_1(x, y, z), f_2(x, y, z), f_3(x, y, z)) = (x^2 + y^2)\hat{\mathbf{i}} + (y^2 + z^2)\hat{\mathbf{j}} + (z^2 + x^2)\hat{\mathbf{k}}$.

2p **5a** Calculate $\nabla \cdot \mathbf{f}$.

2p **5b** Calculate $\nabla \times \mathbf{f}$.

6p **5c** Evaluate/calculate the following quantities (2p each):
i) $\nabla \cdot (\nabla \times \mathbf{f})$
ii) $\nabla(\nabla \cdot \mathbf{f})$
iii) $\nabla \times (\nabla \times \mathbf{f})$

Extra Space

6 You can write your answers in the extra space here if you exceed the allotted space